MEETMIND – AI-ASSISTED MEETING MINUTES AND ACTION TRACKER

A PROJECT REPORT

Submitted by

SANSHIA J

(2116240701471)

in partial fulfilment of the End Semester Examination for the course

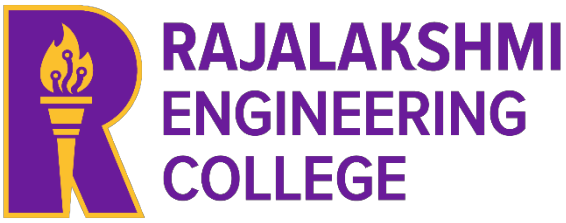
CB23F35

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “MeetMind – AI-Assisted Meeting Minutes and Action Tracker” is the bonafide work of SANSHIA J (2116240701471), who carried out the project work as a member of a project team on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SANSHIA J (2116240701471), hereby declare that the project report titled "MeetMind – AI-Assisted Meeting Minutes and Action Tracker", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of work carried out by me as a member of a project team. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts and workflows — was designed and developed by our team, and this report was prepared by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

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SANSHIA J

ABSTRACT

Teams in colleges and workplaces hold regular meetings to plan work and review progress, but the details of these meetings are rarely recorded in one place. Meetings are announced through e-mail and chat, notes are kept on paper or in personal files, minutes are typed late or not at all, and the tasks agreed in a meeting often have no recorded owner, due date or status. The team lead has no reliable view of which decisions were taken and which tasks are still pending.

This project presents MeetMind, an AI-assisted meeting minutes and action tracker built on Zoho Creator using a low-code approach. The application has four linked forms —Meetings, Meeting_Minutes, Action_Items and Team_Members—with a list report for each. A Zia Keyword Extraction field automatically extracts the key topics of each set of meeting notes. Six form workflows written in Deluge record a confirmation message in the workflow log whenever a record is added or edited, and two scheduled workflows are configured to e-mail meeting reminders but are currently inactive. A User Engagement Dashboard page, built with the UI Builder, shows meeting and action-item counts, charts by status and by team member, a status board and embedded reports. The application also contains a sample Data Source form created by the platform, which is not part of the MeetMind modules.

Testing was limited to basic manual testing: records were entered through every form and the reports, the dashboard and the extracted keywords were checked against the entered data. The dashboard counts of meetings and action items matched the reports, and keywords were extracted for both test sets of minutes. A review of the exported application definition identified six open defects, including dashboard panels without filtering criteria, a chart bound to the sample form, duplicate workflow triggers and the UTC time zone, which are recorded in the defect log. Integration with Zoho CRM, REST API access, custom permission profiles, Stage and Production deployment and formal user acceptance testing were not implemented and are listed as future enhancements.

Keywords: *Low-Code Development; Zoho Creator; Zia Keyword Extraction; Workflow Automation; Meeting Management; Action Tracking*

PROJECT AT A GLANCE

Attribute	Details
Project Title	MeetMind – AI-Assisted Meeting Minutes and Action Tracker
Application Domain	Workplace productivity – meeting management, minutes and action-item tracking
Student Name & Register No.	SANSHIA J – 2116240701471
Zoho Creator Edition	Zoho Creator trial edition (application header on 09-11-2026: “Trial expires in 12 days”)
Application Name (Link Name)	MeetMind (link name: meetmind)
Published / Live URL	Not available
Demo Credentials (per role)	Not available
Number of Forms / Subforms	5 (4 MeetMind forms + 1 sample form, Data Source) / 0
Number of Reports (by type)	List: 5 (4 MeetMind + 1 sample, All Data Points) | Kanban: 0 | Calendar: 0 | Pivot/Chart: 0 (five charts and one board are embedded in the dashboard page) | Others: 0
Number of Pages / Dashboards	1 – User Engagement Dashboard (UI Builder: KPI panels, five charts, a board and two embedded reports)
Number of Workflows	Form: 6 | Schedule: 2 (inactive) | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	0 functions / 0 custom actions; 6 Deluge scripts (info statements) in form workflows (DS-01 to DS-06)
Integrations	Zoho Mail (send-email action in two schedules, inactive). Zoho CRM and REST API: Not Implemented
Roles & Profiles Defined	Default profiles: Administrator, Read, Write, Customer (customer portal). Role: CEO
Environments Used	Development (Stage and Production: Not Implemented)
Date of Production Deployment	Not available
Source Code Repository	Not available

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script identifier in this report)
ER	Entity–Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
ID	Identifier
IST	Indian Standard Time
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OAuth	Open Authorization
PDF	Portable Document Format
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
UTC	Coordinated Universal Time
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Every organisation, department and student project team depends on meetings to plan work, review progress and take decisions. A meeting is only useful, however, if what was discussed is written down, the decisions are recorded, and each task agreed in the meeting has a clear owner and a due date. In practice these details are often lost: notes are kept in personal notebooks or chat messages, minutes are typed late or not at all, and the people who agreed to do something are reminded only informally.

In most small teams this process is informal. The organiser calls the meeting through e-mail or a messaging group, someone takes notes, and the follow-up depends on the memory and goodwill of individual members. The organiser or team lead has no single place that shows which meetings were held, what was decided, which tasks are still pending and who is responsible for them. The process described in this report is that of a typical project team or small department that meets regularly to review work, and the MeetMind application was designed for that setting.

Three groups of people are involved: the meeting organiser, who schedules meetings and records minutes; team members, who attend meetings and carry out the action items assigned to them; and an administrator, who maintains the application and wants an overview of meetings and task progress. The business process addressed by this project is recording meetings and their minutes, extracting the key topics from the meeting notes, tracking each action item through to completion, and giving the team a dashboard view of this activity.

1.2 Problem Statement

Meeting records and follow-up tasks are currently kept in scattered e-mails, chat messages, paper notes and personal files. There is no single place where the organiser can record a meeting and its minutes, where team members can see the tasks assigned to them, and where the team lead can see the overall status. The result is lost decisions, unclear task ownership, missed deadlines and no measurable view of how many tasks are pending or completed. The specific pain points are:

- Meeting details such as date, time, agenda, location and organiser are announced in separate messages and are not stored in one place.
- Minutes, decisions and summaries are written inconsistently, if at all, and are difficult to search later.

- The main topics of a long set of meeting notes are not identified, so a reader has to go through the full text to find what was discussed.
- Action items are agreed verbally, so the owner, priority, due date and status of a task are often unclear.
- The team lead must count meetings and pending tasks by hand, so status summaries are slow to prepare and quickly become outdated.

1.3 Objectives

- To design a data model of four linked forms for meetings, meeting minutes, action items and team members.
- To use the Zia Keyword Extractor AI field of Zoho Creator to extract key topics automatically from meeting notes.
- To provide list reports and a dashboard page that give a live view of meetings, action items and team members.
- To configure form workflows and scheduled workflows for submission logging and meeting-reminder e-mails using Deluge and built-in workflow actions.
- To record the access settings, testing performed and known defects of the application so that the work remaining before deployment is clearly documented.

1.4 Scope of the Project

In scope: three intended user roles (meeting organiser, team member and administrator); recording meetings with title, date, time, agenda, location, organiser and status; recording meeting minutes linked to a meeting, with notes, decisions and a summary; automatic keyword extraction from the meeting notes using a Zia AI field; recording action items with assignee, due date, status, description, priority and a linked team member; a team member directory; list reports for each module; a dashboard page with counts, charts, a board and embedded reports; six form workflows that write confirmation messages to the log; and two scheduled reminder e-mail workflows, which are configured but inactive. The application runs on the web, with phone and tablet layouts configured in Zoho Creator.

Out of scope: integration with Zoho CRM or other external systems, access through the Zoho Creator REST API, approval processes and Blueprints, custom permission profiles and customer-portal access, deployment to Stage and Production environments, automatic transcription of meetings, and AI-generated meeting summaries. These items are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

Low-code platforms allow applications to be assembled largely through visual builders instead of hand-written code, which shortens development time and lets people without a professional programming background take part in building and maintaining applications [1]. This matched the nature of the MeetMind problem: the data model is small (four linked forms), the processes are routine (schedule, record, assign, track, report), and the application had to be built and adjusted quickly within the duration of the course.

Zoho Creator was chosen because it offers the required building blocks in one hosted service: forms and lookup fields for the data model, reports and pages for the user interface, form and schedule workflows with the Deluge scripting language for automation, ready-made AI fields such as keyword extraction, and built-in e-mail actions [2]. Hosting, backups and layouts for phones and tablets are provided by the platform, so no server had to be set up. The main costs are the Zoho Creator subscription (a trial edition was used for this project) and development time; no separate web server, database or mobile application had to be bought or maintained. Low-code also has limits, which are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement, a meeting is called through e-mail, a chat group or a phone call. During the meeting, someone takes notes on paper or in a personal notebook. Minutes, if they are prepared at all, are typed later and circulated by e-mail. Tasks are agreed verbally, and the owner and due date are often not written down. Before the next meeting, the organiser compiles the status of each task by asking members individually, and the decisions and notes of earlier meetings remain scattered across mailboxes and personal files. Figure 2.1 shows this as-is flow.

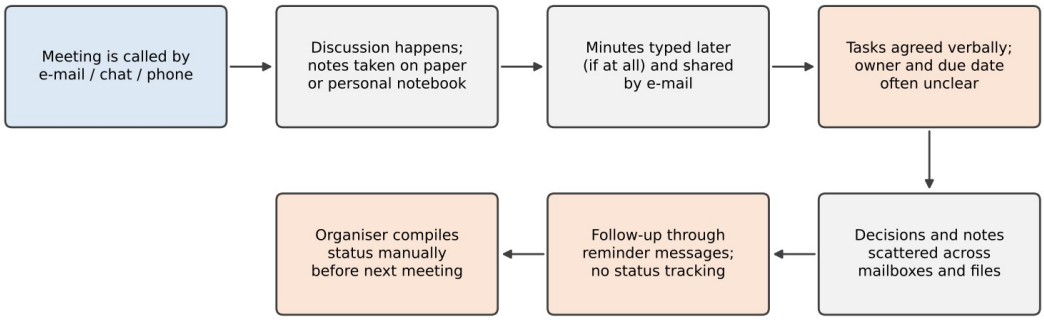


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- There is no central record of meetings, their agenda, location, organiser and status.
- Minutes and decisions are not stored in a consistent structure and cannot be searched easily.
- The key topics of a meeting are not identified without reading the full notes.
- Action items have no recorded owner, priority, due date or status, so follow-up is unreliable.
- Status reporting is manual and delayed, and there is no overview of pending and completed tasks.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Database design, back end, front end, authentication and deployment would be built separately and would take weeks.	Forms, reports, workflows and a dashboard page were configured in visual builders; Deluge was written only for short workflow scripts.
Skills required	Web framework, database, API and DevOps skills.	Basic data modelling and Deluge scripting.
Hosting & infrastructure	Server, database and SSL certificate to be provisioned and maintained.	Hosted by Zoho Creator; no infrastructure to manage.
Mobile application	Separate mobile app or a responsive front end.	Phone and tablet layouts generated from the same forms and reports.
AI features	A keyword-extraction library or an external AI service would have to be integrated and hosted.	Keyword Extraction is available as a ready-made Zia AI field, configured on the Meeting Notes field.
Security & access control	Login, roles and permissions to be coded and tested.	Zoho account sign-in and default permission profiles are built in.
Maintenance & change	Code changes, redeployment and regression testing.	Fields, reports and workflows edited in the builders; the application can be exported as a .ds file.
Customisation limits	Practically unlimited.	Bound by platform features; the dashboard is limited to the panels, charts, boards and reports offered by the UI Builder (Section 8.4).
Cost	Developer time plus hosting and maintenance.	Platform subscription plus configuration time.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	“Low-Code Platform”, Business & Information Systems Engineering	Defines low-code platforms and discusses citizen development, development productivity and organisational agility.	Supports the choice of a low-code platform for a small team application that has to be built quickly.
2	A. Sahay et al., 2020 [3]	Supporting the understanding and comparison of low-code development platforms	Surveys eight representative platforms with a conceptual comparison framework to help choose the platform that fits given requirements.	Guides the selection criteria used for Zoho Creator: forms, workflows, AI fields and hosting.

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	D. Di Ruscio et al., 2022 [4]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines how low-code development relates to model-driven engineering, both being based on abstraction.	Explains why a data model (ER design) comes before form building in this project.
4	R. Waszkowski, 2019 [5]	Low-code platform for automating business processes in manufacturing	Reports the automation of business processes with a low-code platform in a manufacturing setting.	Shows that routine business processes can be automated, as MeetMind structures meeting and task follow-up.
5	Zoho Corporation, 2026 [6]	Zoho Creator help: Keyword Extraction field	Describes the Keyword Extraction AI field, which extracts keywords from the text of a chosen base field of the same form.	Basis of the Extracted_Keywords field on the Meeting_Minutes form (Section 5.9).
6	Zoho Corporation, 2026 [7], [8]	Zoho Creator help: Schedules and the send-email action in schedules	Explains form-based and custom schedules and the send-email action that can run when a schedule starts.	Basis of the two Upcoming_Meeting_Reminder schedules (Section 5.6).

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development influenced the design of MeetMind. Citizen development means that people who are not professional programmers, such as a team lead or a department coordinator, can build and maintain applications; the visual builders of Zoho Creator and the small number of scripts in this project were chosen with that in mind [1]. Composable applications are assembled from existing services rather than built from scratch, which is why e-mail reminders use the built-in Zoho Mail action instead of a custom mail service. Hyper-automation combines workflows, integrations and AI to automate a process end to end; the logging workflows and the reminder schedules of this project are only a first step in that direction.

Platform vendors, including Zoho, are also adding AI assistance to application building and to data fields. MeetMind uses one such capability, the Zia Keyword Extraction field, which turns free-text meeting notes into a list of key topics without any code. Surveys of the platform landscape show how quickly such capabilities are changing [3], and low-code is increasingly discussed alongside model-driven engineering [4]. AI-generated meeting summaries and automatic creation of action items from the minutes are therefore recorded as the main future enhancements of this project.

2.6 Proposed System

MeetMind is a Zoho Creator application that addresses each limitation listed in Section 2.2. A central record of meetings is provided by the Meetings form and the All Meetings report, which hold the title, date, time, agenda, location, organiser and status of each meeting. Minutes are recorded in the Meeting_Minutes form, which must be linked to a meeting and stores the notes, the decisions taken and a summary in a consistent structure. A Zia Keyword Extractor field fills in the key topics of each set of notes automatically. Action items are recorded in the Action_Items form with an assignee, a due date, a status, a priority and a link to a team member. The User Engagement Dashboard page replaces manual counting with live counts, charts, a status board and embedded reports. Finally, Zoho account sign-in and the default permission profiles of the platform control who can use the application; custom role-based access is listed as future work.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1 lists the intended user roles. In the current version of the application these roles are not separated by user accounts or custom profiles; all records were entered under the application owner's account (Section 5.8).

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Meeting Organiser	A team lead or member who calls meetings and records what was discussed.	Schedule meetings; record minutes, decisions and summaries; create action items.
Team Member	A member of the team who attends meetings and carries out assigned tasks.	View meetings and minutes; update the status of the action items assigned to them.
Administrator	The developer or owner responsible for the application.	Maintain team member records and all data; view the dashboard; manage workflows, schedules and profiles.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a meeting to be recorded with title, date, time, agenda, location, organiser and status (Scheduled, In Progress, Completed or Cancelled).	Meeting Organiser	High
FR-02	The system shall allow minutes to be recorded for a meeting selected from the existing meetings, with meeting notes, decisions taken and a meeting summary.	Meeting Organiser	High
FR-03	The system shall extract keywords automatically from the meeting notes and store them with the minutes.	System	Medium
FR-04	The system shall allow an action item to be recorded with title, assignee, due date, status, description, priority and a linked team member.	Meeting Organiser, Team Member	High
FR-05	The system shall allow team members to be registered with name, e-mail, role, department and status.	Administrator	Medium
FR-06	The system shall provide a list report for each module, from which records can be viewed, edited, duplicated and deleted.	Administrator	High

FR ID	Requirement	Role(s)	Priority
FR-07	The system shall display a dashboard with meeting and action-item counts, charts of action items and meetings by status, a status board and embedded reports.	Administrator	Medium
FR-08	The system shall record a confirmation message in the workflow log whenever a record is added or edited.	System	Low
FR-09	The system shall be able to send a reminder e-mail about upcoming meetings on a schedule.	System	Medium
FR-10	The system shall limit what each user can do according to the permission profile assigned to the user.	Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard page should load without noticeable delay for the records of a small team.
NFR-02	Security	Only users signed in with a Zoho account and shared on the application should be able to use it, within the permissions of their profile.
NFR-03	Usability	The application shall be usable in a web browser and, through the phone and tablet layouts, on mobile devices.
NFR-04	Availability	The application shall be hosted on Zoho Creator's cloud, so that no self-managed server is needed.
NFR-05	Maintainability	Components should follow a consistent naming convention and the application shall be exportable as a .ds file for backup.
NFR-06	Auditability	Changes to records should be traceable through the record-history features of Zoho Creator.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition), India data centre (zoho.in)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used (REST API access not implemented)
Other Zoho services	Zia AI (Keyword Extraction field); Zoho Mail (send-email action in schedules, currently inactive)
Mobile	Phone (one-column) and tablet (sliding-pane) layouts configured in the application

Category	Specification
Client hardware	Windows laptop or desktop with a current web browser and an internet connection

3.5 Use Case Model

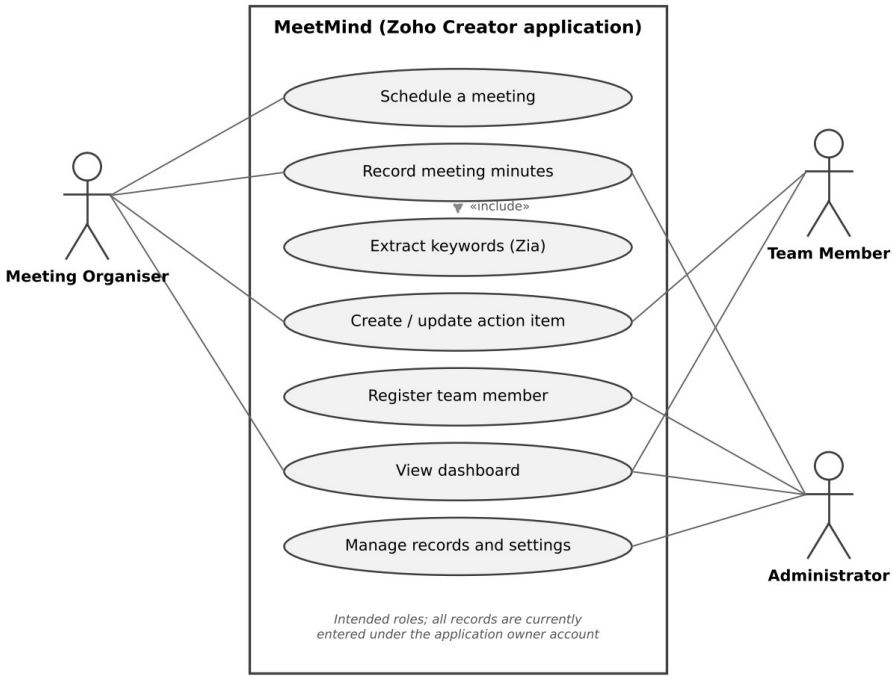


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Record Minutes for a Meeting

Use Case ID	UC-01
Use Case Name	Record minutes for a meeting
Actor(s)	Meeting Organiser
Pre-condition	The meeting has been recorded in the Meetings form.
Main Flow	1. The organiser opens the Meeting_Minutes form. 2. The organiser selects the meeting in the Meetings lookup. 3. The organiser enters the meeting title, meeting date, meeting notes, decisions taken and summary. 4. The organiser clicks Submit. 5. The system saves the record and shows the success message. 6. The Zia Keyword Extractor fills the Extracted_Keywords field from the meeting notes. 7. The Meeting_Minutes_Submission workflow writes two info messages to the log.
Alternate Flow	No meeting is selected: the form does not save the record because the Meetings lookup is mandatory.

Use Case ID	UC-01
Post-condition	The minutes are stored, linked to the meeting, with extracted keywords.
Related FR	FR-02, FR-03, FR-08

Table 3.6: Use Case Description – Create and Track an Action Item

Use Case ID	UC-02
Use Case Name	Create and track an action item
Actor(s)	Meeting Organiser, Team Member
Pre-condition	The team member who will carry out the task is registered in the Team_Members form.
Main Flow	1. The user opens the Action_Items form. 2. The user enters the action title, assignee, due date, action status, description and priority, and selects the team member. 3. The user clicks Submit and the record is saved. 4. Later, the user opens the record from the Action_Items Report and updates the Action Status. 5. The dashboard shows the updated counts the next time the page is loaded.
Alternate Flow	The user leaves the Team_Members field empty: the record is still saved, and the item appears as “Unknown” in the Action Items by Team Member chart (BUG-04).
Post-condition	The action item is stored with its current status and appears in the report, charts and board.
Related FR	FR-04, FR-06, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

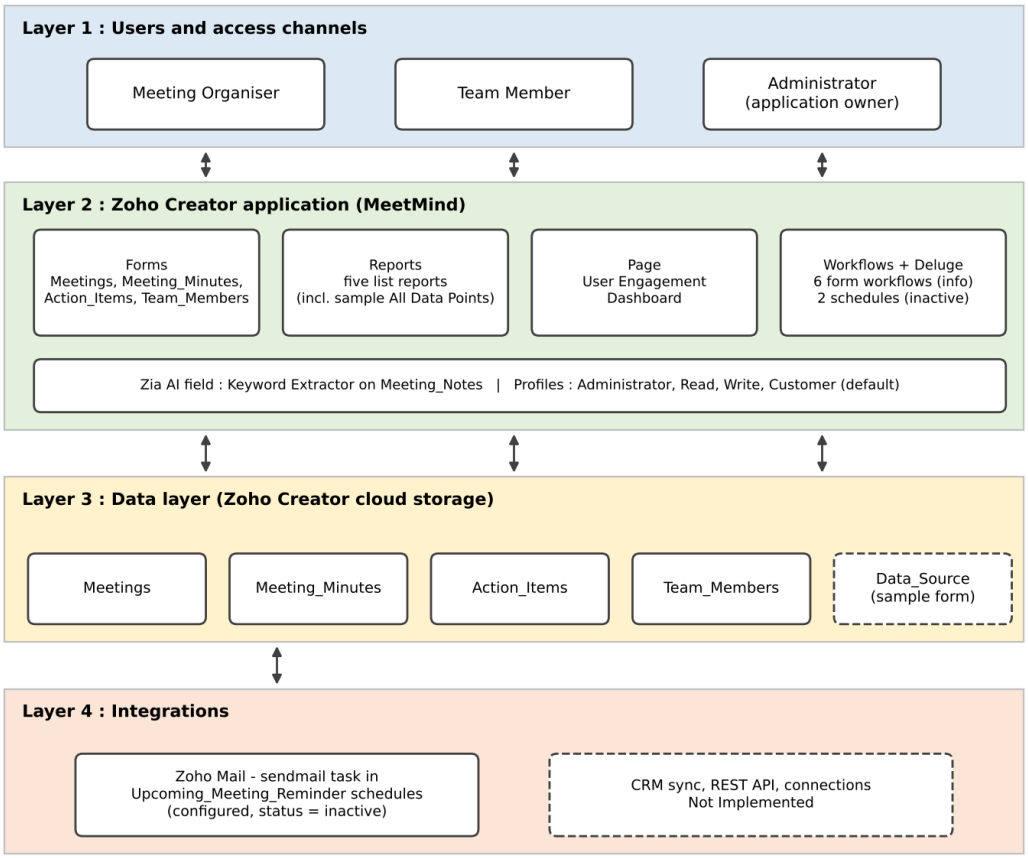


Figure 4.1: System Architecture of the Application

The application follows a four-layer architecture. In the first layer, the intended users—meeting organisers, team members and the administrator—reach the application through a web browser or the phone and tablet layouts after signing in with a Zoho account. In the second layer, the Zoho Creator application contains the forms that capture data, the reports that display it, the User Engagement Dashboard page, and the form and schedule workflows that automate behaviour; the Zia Keyword Extractor field and the default permission profiles also belong to this layer.

The third layer is the data layer, where Zoho Creator stores the records of the four MeetMind forms—Meetings, Meeting_Minutes, Action_Items and Team_Members—together with the sample Data_Source form that was created with the application and is not part of the MeetMind modules. The fourth layer holds the integrations: the only integration configured is Zoho Mail, used by the send-email action of the two reminder schedules, which are currently inactive. Zoho CRM synchronisation, REST API access and external connections are not implemented.

4.2 Data Model

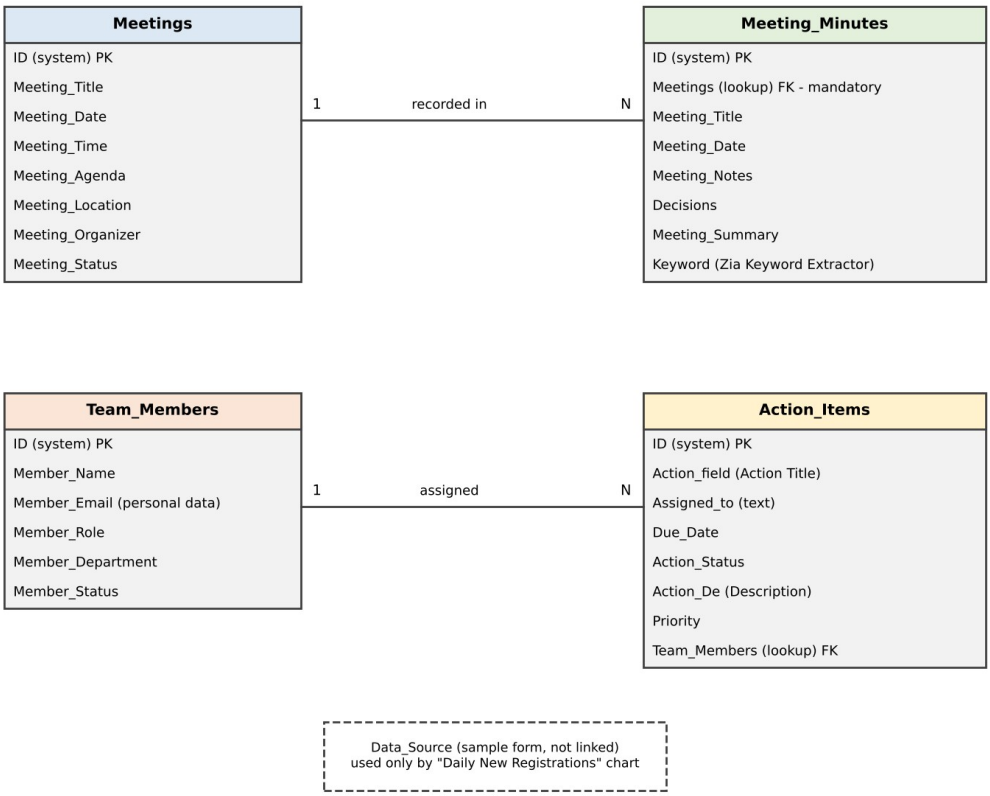


Figure 4.2: Entity–Relationship Diagram

The data model has four MeetMind entities. A meeting can have many sets of minutes and a team member can be linked to many action items, so Meeting_Minutes holds a lookup to Meetings and Action_Items holds a lookup to Team_Members (one-to-many in both cases). In Zoho Creator these relationships are realised through lookup fields, which store the system-generated record ID of the parent record and display a chosen field, here the meeting title and the member name. The Meetings lookup in Meeting_Minutes is the only mandatory field in the application. Meeting_Minutes also has its own Meeting_Title and Meeting_Date fields, which are entered by hand and are not filled from the selected meeting, and Action_Items has a free-text Assigned_to field in addition to the Team_Members lookup. No subform is used. The sample Data_Source form is not linked to any MeetMind form. Table 4.1 summarises the entities.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Meetings	Meeting_Title, Meeting_Date, Meeting_Time, Meeting_Agenda, Meeting_Location, Meeting_Organizer, Meeting_Status	Meeting_Minutes	One-to-Many	Lookup in Meeting_Minutes (Meetings), mandatory
Meeting_Minutes	Meetings, Meeting_Title, Meeting_Date, Meeting_Notes, Decisions, Meeting_Summary, Keyword	Meetings	Many-to-One	Lookup field Meetings (displays Meeting_Title)
Team_Members	Member_Name, Member_Email, Member_Role, Member_Department, Member_Status	Action_Items	One-to-Many	Lookup in Action_Items (Team_Members), optional
Action_Items	Action_field, Assigned_to, Due_Date, Action_Status, Action_De, Priority, Team_Members	Team_Members	Many-to-One	Lookup field Team_Members (displays Member_Name)
Data_Source (sample)	Date1, Total_Users, Active_Today, New_Registrations and 10 other metric fields	None	—	Sample form created with the application; used only by one dashboard chart

4.3 Form Design

Table 4.2: Field Specification – Meetings Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Meeting Title	Meeting_Title	Single Line	No	—	Display field of the lookup in Meeting_Minutes
Meeting Date	Meeting_Date	Date	No	dd-MMM-yyyy; all days allowed	Label stored as “Meeting Date”
Meeting Time	Meeting_Time	Time	No	hh:mm:ss	24-hour format
Meeting Agenda	Meeting_Agenda	Multi Line	No	—	Height 100 px
Meeting Location	Meeting_Location	Single Line	No	—	—
Meeting Organizer	Meeting_Organizer	Single Line	No	—	Free text

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Meeting Status	Meeting_Status	Drop Down	No	Scheduled, In Progress, Completed, Cancelled	Used by the Meetings by Status chart

Table 4.3: Field Specification – Meeting_Minutes Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Meetings	Meetings	Lookup (Meetings)	Yes	Shows Meeting_Title; sorted ascending	Only mandatory field in the application
Meeting Title	Meeting_Title	Single Line	No	—	Entered by hand
Meeting Date	Meeting_Date	Date	No	dd-MMM-yyyy	Entered by hand
Meeting Notes	Meeting_Notes	Multi Line	No	—	Base field for keyword extraction
Decisions Taken	Decisions	Multi Line	No	—	—
Meeting Summary	Meeting_Summary	Multi Line	No	—	Height 150 px
Extracted_Keywords	Keyword	AI field – Keyword Extraction	No	Base field: Meeting_Notes	Filled by Zia

Table 4.4: Field Specification – Action_Items Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Action Title	Action_field	Single Line	No	—	Link name is Action_field
Assigned To	Assigned_to	Single Line	No	—	Free text, not linked to Team_Members
Due Date	Due_Date	Date	No	dd-MMM-yyyy	—
Action Status	Action_Status	Drop Down	No	Pending, In Progress, Completed, On Hold	Used by charts, board and KPI
Action Description	Action_De	Multi Line	No	—	Link name is Action_De
Priority	Priority	Drop Down	No	High, Medium, Low	—

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Team_Members	Team_Members	Lookup (Team_Members)	No	Shows Member_Name; sorted ascending	Used by the per-member chart

Table 4.5: Field Specification – Team_Members Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Member Name	Member_Name	Single Line	No	—	Display field of the lookup in Action_Items
Member Email	Member_Email	Email	No	Up to 80 characters	Flagged as personal data
Member Role	Member_Role	Drop Down	No	Admin, Manager, Team Member	—
Member Department	Member_Department	Single Line	No	—	Free text
Member Status	Member_Status	Drop Down	No	Active, Inactive	—

4.4 Process and Workflow Design

The main process is the life cycle of a meeting and its follow-up, shown in Figure 4.3. When a meeting is submitted, two form workflows run on success and write confirmation messages to the log. Minutes are then recorded against the meeting; the Zia Keyword Extractor fills the keywords from the notes and a third workflow writes to the log. Action items are created and updated by the team, and two further workflows log each submission. The dashboard page reads the stored records whenever it is opened. A team member registration triggers its own logging workflow (not shown in the figure).

Two scheduled workflows, both named Upcoming_Meeting_Reminder, were designed to send a reminder e-mail about upcoming meetings at 07:10 UTC. Both are configured with the status inactive, so they do not run at present; this is shown with a dashed border in Figure 4.3. Scripts DS-01 to DS-06 are listed in Section 5.5.

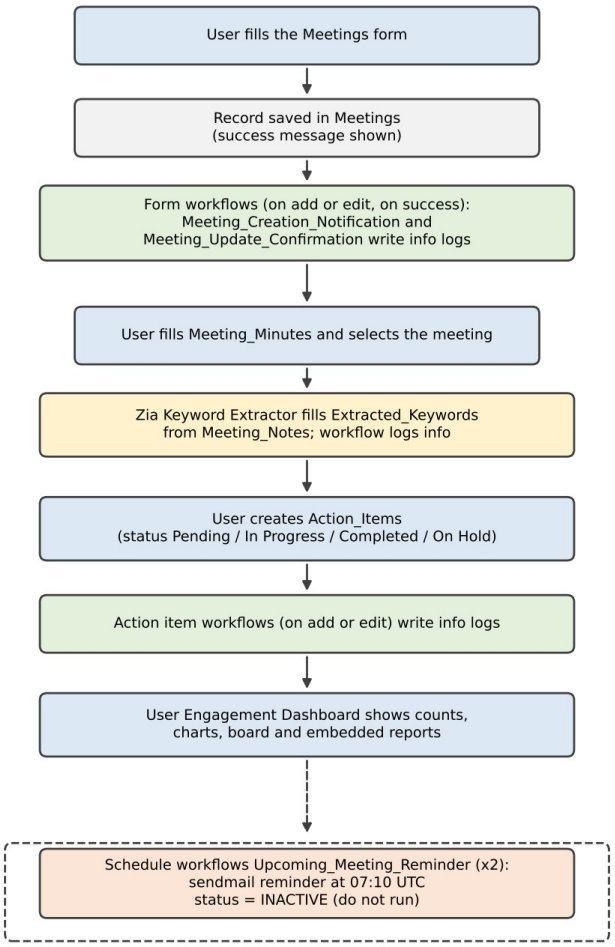


Figure 4.3: Workflow / Process Flow for a Meeting and its Follow-up

4.5 User Interface and Navigation Design

The navigation menu groups each MeetMind form with its report, so that a user finds data entry and the matching list in one section, and places the dashboard page in the Home section. The menu also contains the sample All Data report and Add New form, and the system sections App Preferences, Approvals and Reports. Figure 4.4 shows the menu and the wireframe of the User Engagement Dashboard: a welcome banner, three KPI cards, three column charts, a KPI strip, a pie chart, a per-member column chart, a status board and two embedded reports.

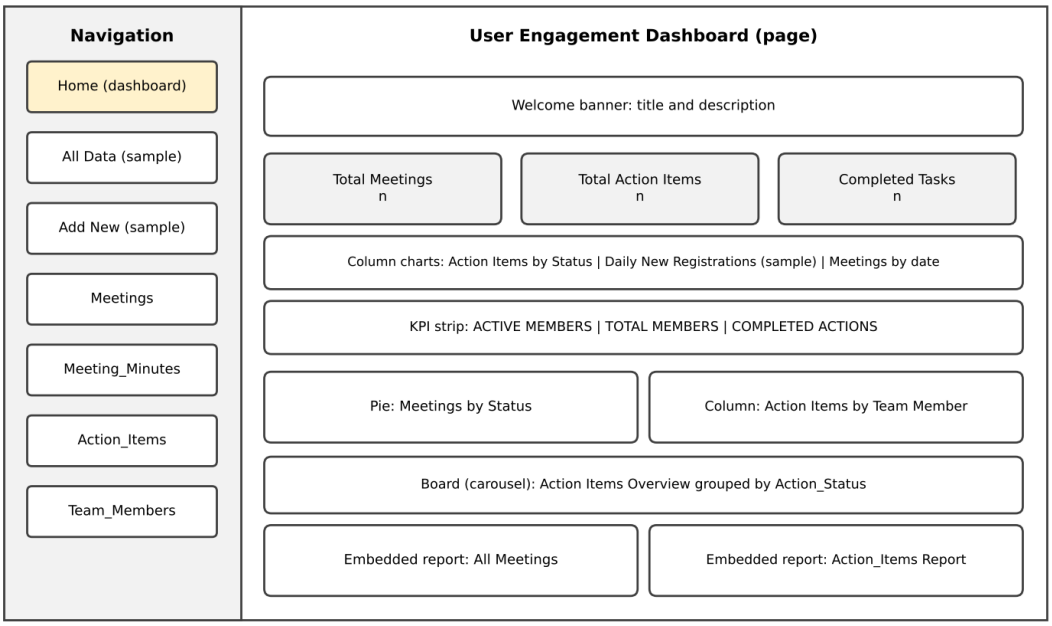


Figure 4.4: Navigation Structure and Wireframe of the User Engagement Dashboard

4.6 Access Control Design

Table 4.6 is the intended permission design for MeetMind. A = Add, E = Edit, V = View, D = Delete, — = no access. This matrix has not been configured in the application: only the default profiles of Zoho Creator exist, and no additional users or custom profiles have been set up. Section 5.8 describes the current configuration.

Table 4.6: Role–Permission Matrix (Design)

Component	Administrator	Meeting Organiser	Team Member	API client
Form: Meetings	A E V D	A E V	V	Not Implemented
Form: Meeting_Minutes	A E V D	A E V	V	Not Implemented
Form: Action_Items	A E V D	A E V	V E (own items)	Not Implemented
Form: Team_Members	A E V D	V	V	Not Implemented
Report: all MeetMind reports	V E D	V E	V	Not Implemented
Page: User Engagement Dashboard	V	V	V	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator workspace on the India data centre with the display name MeetMind and the link name meetmind. The exported application definition (MeetMind.ds, version 1.0, generated on 06-Nov-2026) records the date format dd-MMM-yyyy, the time zone UTC and the 24-hour time format. The application runs on a Zoho Creator trial edition; on 09-11-2026 the application header showed that the trial would expire in 12 days.

The navigation menu has ten sections: Home (the User Engagement Dashboard page), All Data (the sample All Data Points report), Add New (the sample Data Source form), App Preferences (connection and language selection), Approvals (completed and pending tasks), Reports (shared analytics), Meetings, Meeting_Minutes, Action_Items and Team_Members. The last four sections each hold a MeetMind form and its report. In live mode the sidebar shows Home, All Data, Add New and the four MeetMind sections. The web layout uses the Lato font with colour option 11 and shows the application icon, which displays the text “MEE”, on the left; the phone layout uses one column with style 3 and a yellow colour scheme, and the tablet layout uses a sliding pane with style 3 and a green colour scheme.

5.2 Forms and Data Capture

Four MeetMind forms were built. **Meetings** captures the meeting title, date, time, agenda, location, organiser and status. Meeting Date is a Date field in the dd-MMM-yyyy format and Meeting Time is a Time field in the hh:mm:ss format; Meeting Status is a drop-down with the values Scheduled, In Progress, Completed and Cancelled. The form shows the success message “Meeting Added Successfully”.

Meeting_Minutes links each set of minutes to a meeting through the mandatory Meetings lookup, which displays the meeting title. The other fields are Meeting Title, Meeting Date, Meeting Notes, Decisions Taken, Meeting Summary and Extracted_Keywords. Extracted_Keywords is a Zia AI field of type Keyword Extraction whose base field is Meeting Notes (Section 5.9). Meeting Title and Meeting Date are entered by hand and are not copied from the selected meeting, which allowed the inconsistency recorded as BUG-05.

Action_Items captures the action title, assigned to (free text), due date, action status (Pending, In Progress, Completed or On Hold), action description, priority (High, Medium or Low) and a Team_Members lookup that displays the member name.

Team_Members captures the member name, e-mail, role (Admin, Manager or Team Member), department and status (Active or Inactive); the e-mail field is flagged as personal data. All four forms use Submit for adding and Update and Cancel for editing, and place labels to the left in the web layout. The only field-level rule is the mandatory Meetings lookup; subforms and layout rules were not used. Field details are in Tables 4.2 to 4.5.

The application also contains a fifth form, **Data Source**, which is a sample form created by Zoho Creator together with the application. It holds a "Creator tip" note and fourteen business-metric fields such as Date, Total Users, Active Today, New Registrations, Revenue, Server Load and User Satisfaction, and is listed in the All Data Points report. It is not part of the MeetMind modules and is not linked to any MeetMind form; its only connection to MeetMind is that the dashboard chart "Daily New Registrations" reads from it (BUG-02).

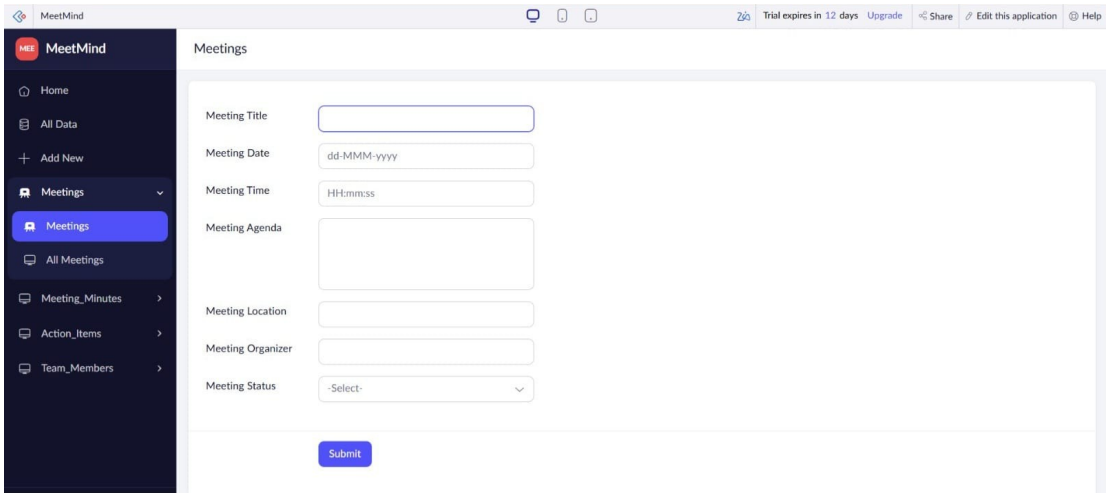


Figure 5.1: Meetings Form in Live Mode

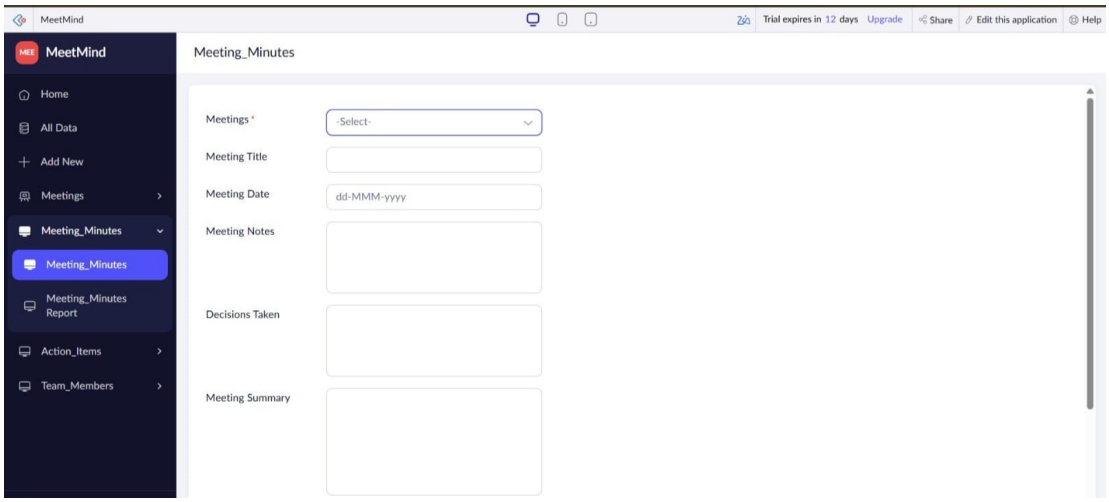


Figure 5.2: Meeting_Minutes Form in Live Mode

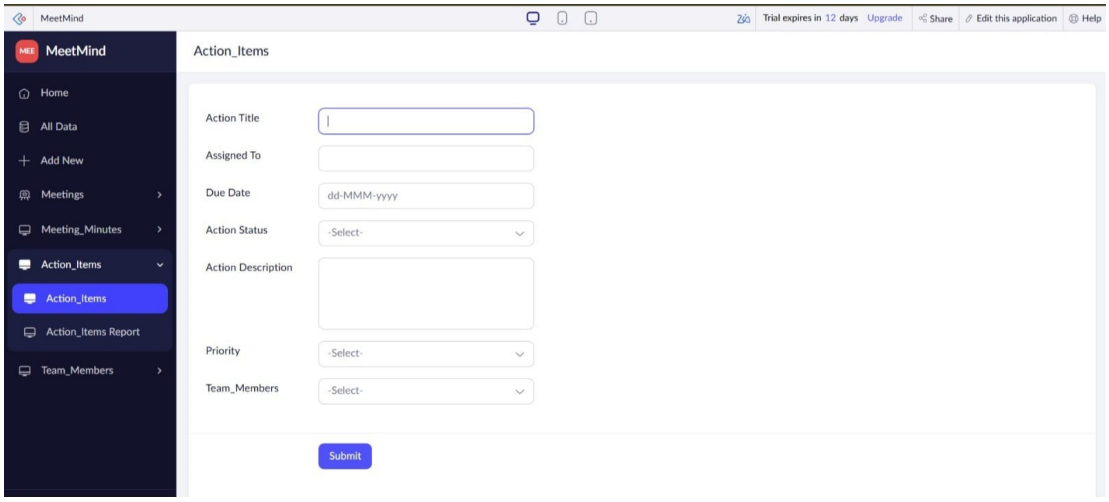


Figure 5.3: Action_Items Form in Live Mode

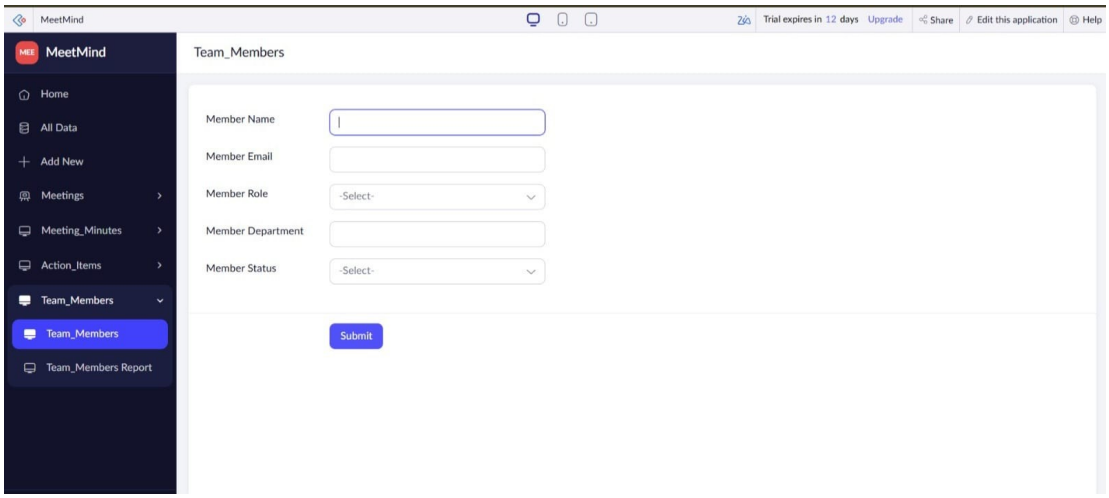


Figure 5.4: Team_Members Form in Live Mode

5.3 Reports

Five list reports exist, one for each form. Each report shows all rows of its form; the quick view and the detail view show the same fields, a click on a row opens the record, and the record menu and the right-click menu offer Edit, Duplicate and Delete. The sample All Data Points report additionally offers Add, Print, Import and Export in its header menu. No filters, grouping, sorting rules or custom actions were configured, and no Kanban, calendar or pivot report was created. Table 5.1 lists the reports; Figures 5.5 to 5.7 show the MeetMind reports with the test records entered during development.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Meetings	List	Meetings	Show all rows; 7 fields; quick and detail views	None (Edit, Duplicate, Delete in record menu)	Application owner

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Meeting_Minutes Report	List	Meeting_Minutes	Show all rows; 7 fields; quick and detail views	None (Edit, Duplicate, Delete in record menu)	Application owner
Action_Items Report	List	Action_Items	Show all rows; 7 fields; quick and detail views	None (Edit, Duplicate, Delete in record menu)	Application owner
Team_Members Report	List	Team_Members	Show all rows; 5 fields; quick and detail views	None (Edit, Duplicate, Delete in record menu)	Application owner
All Data Points (sample)	List	Data_Source	Show all rows; 14 fields; fixed column widths	None (Add, Print, Import, Export in header)	Application owner

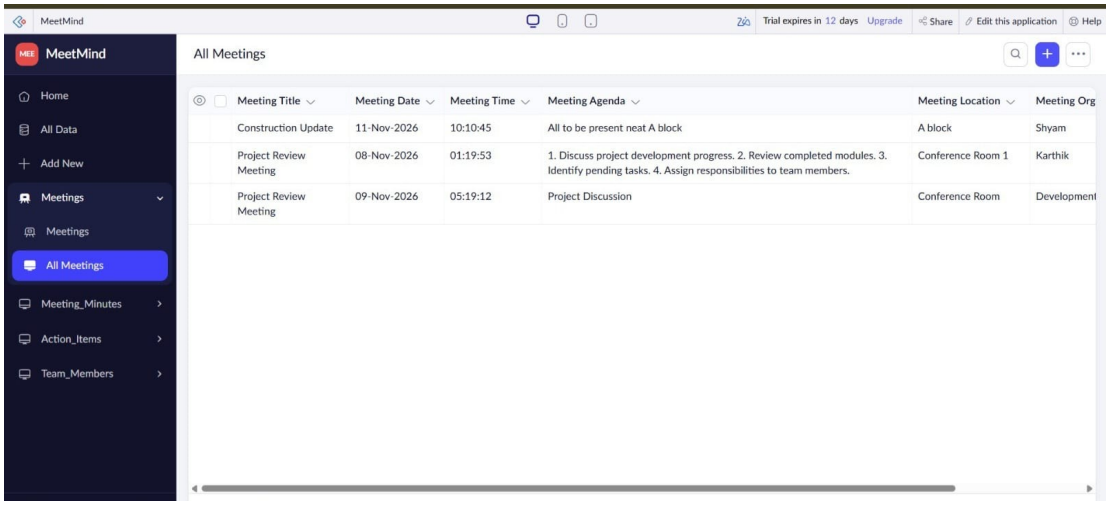


Figure 5.5: All Meetings Report (List View)

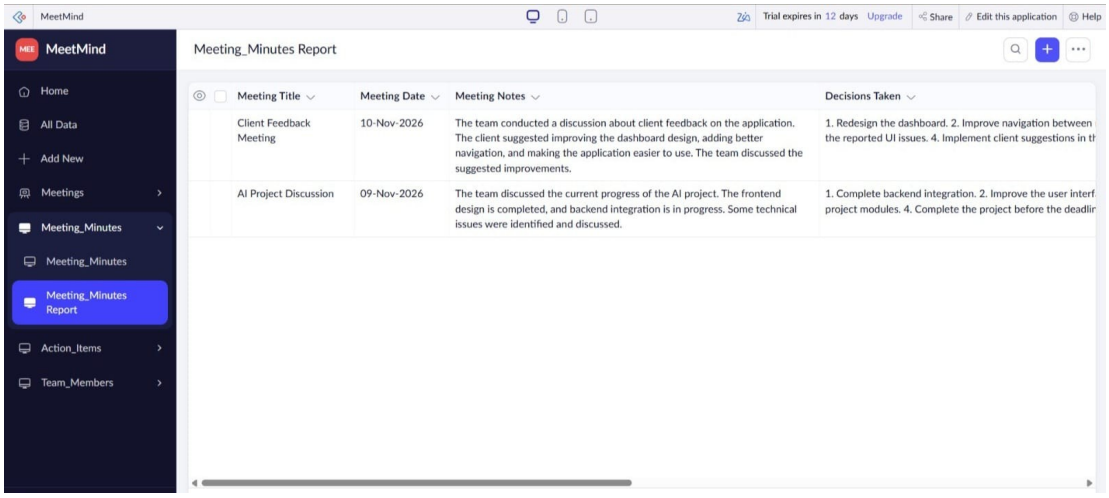


Figure 5.6: Meeting_Minutes Report (List View)

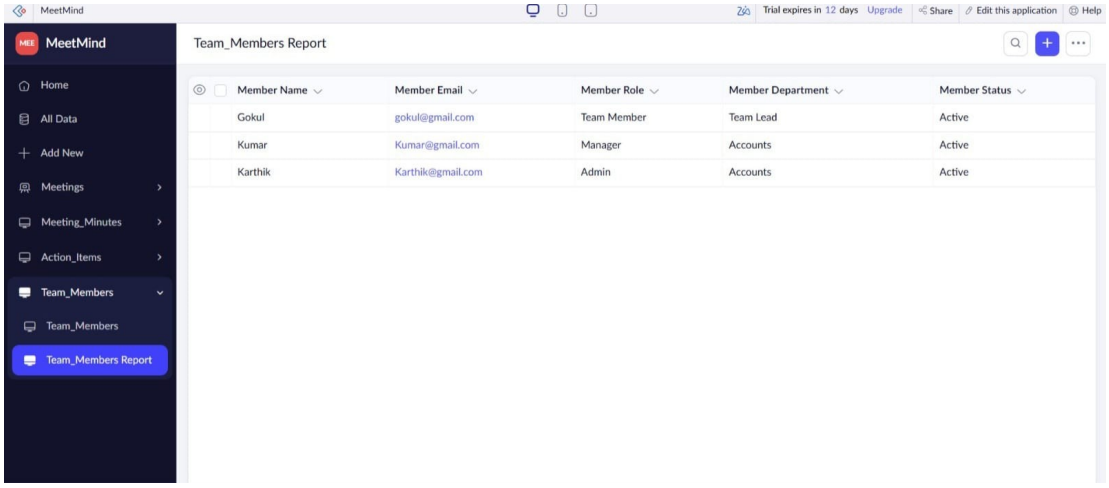


Figure 5.7: Team_Members Report (List View)

5.4 Pages and Dashboards

The User Engagement Dashboard page is built with the UI Builder and is the Home section of the application. Its first panel is a welcome banner with the title “Welcome to MeetMind — Meeting Management Dashboard” and the line “Track meetings, manage action items, and monitor team progress in one place.” The second row holds three KPI cards that use form-data aggregates: Total Meetings (count of Meetings records), Total Action Items (count of Action_Items records) and Completed Tasks (count of Action_Items records with the criteria Action_Status == “Completed”). No Deluge code runs on the page; every value is computed by the page components.

Below the cards are three full-width column charts: Action Items by Status (count of Action_Status values), Daily New Registrations (New_Registrations by Date from the sample Data_Source form) and an untitled chart of meetings by Meeting_Date. A KPI strip then shows ACTIVE MEMBERS, TOTAL MEMBERS and COMPLETED ACTIONS; the first two use the same expression, a count of Member_Name, and the third counts all Action_field values without a status criterion (BUG-01). The next row holds a pie chart of Meetings by Status and a column chart of Action Items by Team Member, followed by a carousel board, “Action Items Overview”, that groups action items by Action_Status, and finally the All Meetings and Action_Items Report reports embedded side by side.

With the test data shown in Figure 5.8, the page reports 3 meetings, 2 action items and 0 completed tasks, and the Action Items by Status chart shows one Pending and one In Progress item. In Figure 5.9 the board shows the In Progress and Pending groups, and the Action Items by Team Member chart shows “Unknown” because the Team_Members lookup was left empty in both action items (BUG-04).

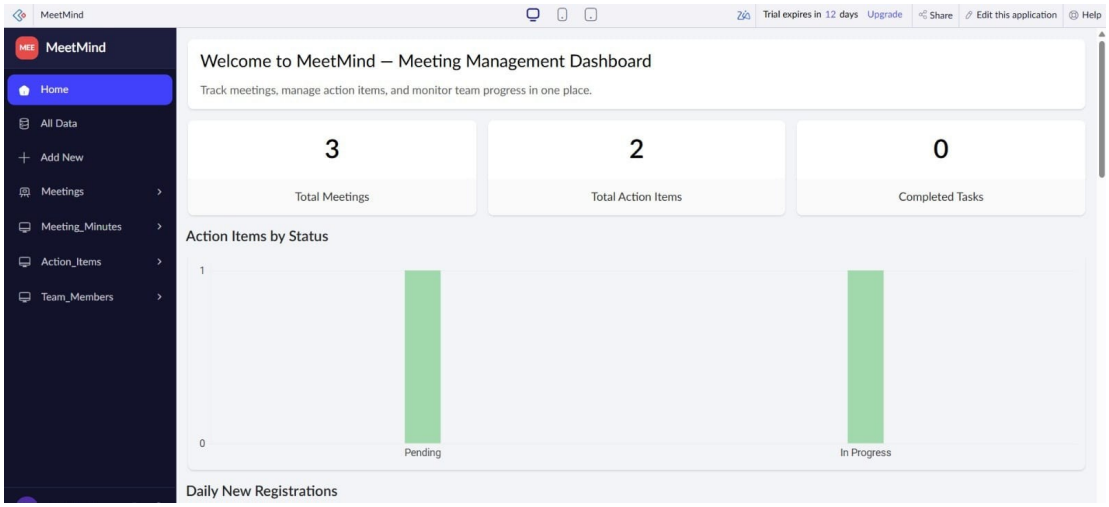


Figure 5.8: User Engagement Dashboard – Banner, KPI Cards and Status Chart

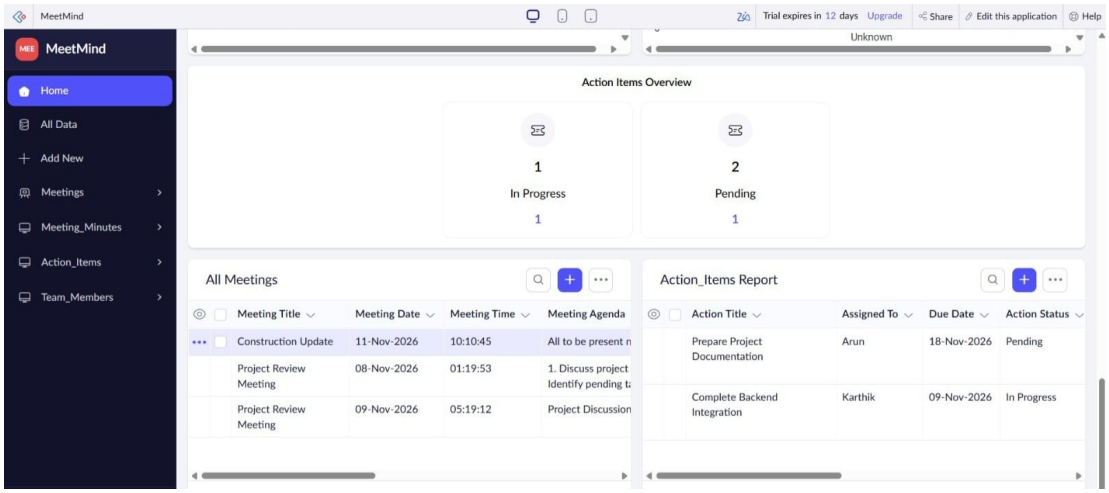


Figure 5.9: User Engagement Dashboard – Status Board and Embedded Reports

5.5 Deluge Scripting

Six Deluge scripts are used in the application. Each one runs in a form workflow on the On Success event and contains two info statements, which write messages to the workflow log; none of them reads, changes or creates records. No Deluge functions, custom actions, page scripts or integration tasks were written. Table 5.2 lists the scripts and the key listings follow.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Meetings → workflow Meeting_Creation_Notification, On Add or Edit → On Success	Log that the meeting form was submitted	info statement
DS-02	Meetings → workflow Meeting_Update_Confirmation, On Add or Edit → On Success	Log that the meeting record was updated	info statement

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-03	Meeting_Minutes → workflow Meeting_Minutes_Submission, On Add or Edit → On Success	Log that the minutes were submitted	info statement
DS-04	Action_Items → workflow Action_Item_Creation_Notification, On Add or Edit → On Success	Log that an action item was submitted	info statement
DS-05	Action_Items → workflow Action_Item_Status_Update, On Add or Edit → On Success	Log that an action item was updated	info statement
DS-06	Team_Members → workflow Team_Member_Registration, On Add or Edit → On Success	Log that a team member was registered	info statement

Code 5.1: DS-01 and DS-02 – Meetings form workflows

```
// DS-01 | Form: Meetings | Workflow: Meeting_Creation_Notification
// Event: On Add or Edit -> On Success
info "MeetMind: Meeting form submitted successfully.";
info "Meeting details have been saved.";

// DS-02 | Form: Meetings | Workflow: Meeting_Update_Confirmation
// Event: On Add or Edit -> On Success
info "MeetMind: Meeting record updated successfully.";
info "Meeting changes have been saved.";
```

Code 5.2: DS-03 – Meeting_Minutes form workflow

```
// DS-03 | Form: Meeting_Minutes | Workflow: Meeting_Minutes_Submission
// Event: On Add or Edit -> On Success
info "MeetMind: Meeting minutes submitted successfully.";
info "Meeting minutes record has been saved.";
```

Code 5.3: DS-04 and DS-05 – Action_Items form workflows

```
// DS-04 | Form: Action_Items | Workflow: Action_Item_Creation_Notification
info "MeetMind: New action item submitted successfully.";
info "Action item has been recorded.";

// DS-05 | Form: Action_Items | Workflow: Action_Item_Status_Update
info "MeetMind: Action item record updated successfully.";
info "Please verify the latest task status.";
```

Code 5.4: DS-06 – Team_Members form workflow

```
// DS-06 | Form: Team_Members | Workflow: Team_Member_Registration
info "MeetMind: New team member registered successfully.";
info "Team member record has been saved.";
```

The info task writes a message to the log of the workflow, which a user can open through the “View Log Details” link that appears on the form after submission. Because every workflow is triggered on add or edit, the “creation” and “update” workflows on the same form run together for every submission: DS-01 and DS-02 both run whenever a meeting is added or edited, and DS-04 and DS-05 both run for every action item. The messages therefore do not distinguish a new record from an edited one (BUG-03). The scripts are kept short and serve as placeholders for later automation such as assignee notifications.

5.6 Workflows and Automation

Eight workflows are configured, listed in Table 5.3. Six are form workflows that run on success when a record is added or edited and execute the scripts DS-01 to DS-06. Two are schedule workflows, both with the display name Upcoming_Meeting_Reminder, that use the built-in send-email action [7], [8]. The first is a form-based schedule on the Meetings form that starts on each record’s Meeting_Date at 07:10:00 UTC, ends on the same date, runs daily, and attaches the All Meetings report as a PDF. The second is a custom schedule that starts on 05-Nov-2026 at 07:10:00 UTC and runs daily without an attachment. Both send the e-mail “Upcoming MeetMind Meeting” from the administrator’s address to a team member’s college e-mail address. **Both schedules have the status inactive, so no reminder e-mail is sent by the application at present.** No approval, Blueprint or batch workflow was configured.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Meeting_Creation_Notification	Form workflow	Meetings: On Add or Edit → On Success	None	Write two info messages to the log	DS-01
Meeting_Update_Confirmation	Form workflow	Meetings: On Add or Edit → On Success	None	Write two info messages to the log	DS-02
Meeting_Minutes_Submission	Form workflow	Meeting_Minutes: On Add or Edit → On Success	None	Write two info messages to the log	DS-03
Action_Item_Creation_Notification	Form workflow	Action_Items: On Add or Edit → On Success	None	Write two info messages to the log	DS-04
Action_Item_Status_Update	Form workflow	Action_Items: On Add or Edit → On Success	None	Write two info messages to the log	DS-05
Team_Member_Registration	Form workflow	Team_Members: On Add or Edit → On Success	None	Write two info messages to the log	DS-06

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Upcoming_Meeting_ Reminder	Schedule (form-based, Meetings) – INACTIVE	Meeting_Date at 07:10 UTC; daily; ends on Meeting_Date	None	Send e-mail with All Meetings report as PDF	App. C.7
Upcoming_Meeting_ Reminder (link name …Reminder1)	Schedule (custom) – INACTIVE	From 05-Nov-2026 07:10 UTC; daily	None	Send reminder e-mail	App. C.8

5.7 Integrations

Table 5.4 lists the integrations. The only Zoho service used by the workflows is Zoho Mail, through the send-email action of the two reminder schedules; because both schedules are inactive, this integration is configured but not in use, and the delivery of reminder e-mails has not been verified. The menu contains the standard Connection component under App Preferences, but no connection is defined in the exported application. Integration with Zoho CRM, calls to external services through invokeurl, and access to application data through the Zoho Creator REST API with OAuth 2.0 were not implemented, so no Postman tests were carried out.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
E-mail reminder	Zoho Mail	Built-in send-email action in two schedules (status inactive)	Zoho account (sender: zoho.adminuserid)	Reminder about upcoming meetings; All Meetings report as PDF (first schedule)
CRM sync	Zoho CRM	N / Not Implemented	—	—
Connections / invokeurl	External services	N / Not Implemented	—	—
Creator REST API	Zoho Creator REST API v2.1	N / Not Implemented (no Postman testing)	OAuth 2.0 not configured	—

5.8 User Management and Role-Based Access

The application uses the default permission profiles that Zoho Creator creates with every application. The exported definition contains three internal profiles—Administrator (all permissions), Read (read permission for all components) and Write (write permission for all components)—and one customer-portal profile, Customer, which by default allows only add and view. One role, CEO, is defined for access to the data of all other users. The Member_Email field is flagged as personal data.

No custom profiles, record-level criteria or customer-portal users have been configured, and no additional users are recorded in the exported definition, so all test records were entered by the application owner. The design in Table 4.6 is therefore not yet implemented and is listed as N / Not Implemented in Appendix A.

5.9 AI and Analytics Features

MeetMind uses one AI feature of Zoho Creator: the **Zia Keyword Extraction** field. The Extracted_Keywords field (link name Keyword) of the Meeting_Minutes form is an AI field of type Keyword Extractor with Meeting_Notes as its base field [6]. When minutes are saved, Zia reads the meeting notes and fills the field with a comma-separated list of key terms. In the two test records, the notes on client feedback produced the keywords “team, application, dashboard design, client feedback, client, navigation, discussion, improvements”, and the notes on the AI project produced “progress, AI project, frontend design, backend integration, team, issues” (Figure 8.2). The field is read-only for the user and needs no code.

Analytics are provided by the User Engagement Dashboard (Section 5.4): three KPI cards, a KPI strip, four column charts, a pie chart and a status board, all built with the UI Builder components. Zoho Analytics was not integrated, and other AI fields such as sentiment analysis or prediction were not used. AI-generated meeting summaries and automatic creation of action items from the minutes are planned as future enhancements.

5.10 Pro-Code Extension

A pro-code extension such as Zoho Catalyst, a JavaScript widget or an external program calling the REST API was not used, because the current requirements—forms, reports, a dashboard, logging workflows and keyword extraction—were met with the visual builders, Zia and short Deluge scripts. Catalyst could later host a service that generates meeting summaries or detects action items in the notes, and a widget could replace the standard charts with an interactive task timeline.

5.11 Mobile and Portal Access

The application has phone and tablet layouts: the phone layout uses one column with style 3 and a yellow colour scheme, and the tablet layout a sliding pane with style 3 and a green colour scheme; labels are placed automatically on both. The layouts were configured but not tested on a phone or tablet. Apart from the default Customer profile, no customer-portal configuration exists, so portal access is not implemented.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Only basic manual testing was performed. Sample records were entered through each MeetMind form in live mode, and the resulting reports and the dashboard page were checked against the entered data. While this report was being prepared, the exported application definition (MeetMind.ds) was also reviewed, which revealed the configuration defects listed in Table 6.2. Automated tests, role-based tests with separate user logins, integration and API tests, and formal user acceptance testing were not carried out. Table 6.1 lists the test cases; a status of Pass or Fail is given only where the result is visible in the screenshots, and cells marked [To be recorded] are left for the student to complete.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Record a meeting	Fill the Meetings form and submit	Record saved and listed in All Meetings	Three meetings listed (Figure 5.5)	Pass
TC-02	FR-02	Record minutes for a meeting	Select a meeting, enter notes, decisions and summary; submit	Record saved and listed in the report	Two minutes records listed (Figure 5.6)	Pass
TC-03	FR-02	Mandatory meeting lookup	Submit minutes without selecting a meeting	Error shown; record not saved	[To be recorded]	Not recorded
TC-04	FR-03	Keyword extraction	Save minutes with meeting notes	Extracted_Keywords filled automatically	Keywords shown for both records (Figure 8.2)	Pass
TC-05	FR-04	Record an action item	Fill the Action_Items form and submit	Record saved and listed in the report	Two action items listed (Figure 8.3)	Pass
TC-06	FR-05	Register a team member	Fill the Team_Members form and submit	Record saved and listed in the report	Three members listed (Figure 5.7)	Pass
TC-07	FR-07	Dashboard KPI cards	Open the dashboard	Counts equal the report counts	3 meetings, 2 action items, 0 completed (Figure 5.8)	Pass
TC-08	FR-07	Action Items by Status chart	Open the dashboard	One bar per status in use	Pending 1, In Progress 1 (Figure 5.8)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-07	KPI strip: Active Members and Completed Actions	Review the panel configuration	Counts use Member_Status and Action_Status criteria	Both panels count all records (BUG-01)	Fail
TC-10	FR-07	Action Items by Team Member chart	Open the dashboard	One bar per team member	Bar labelled “Unknown” (Figure 5.9, BUG-04)	Fail
TC-11	FR-02	Minutes linked to the correct meeting	Compare the Meetings lookup with the meeting title	Lookup matches the minutes	Client Feedback Meeting linked to Construction Update (BUG-05)	Fail
TC-12	FR-08	Workflow log message	Submit a meeting; open View Log Details	Two info messages in the log	“View Log Details” link shown after submission (Figure 8.1); log content [To be recorded]	Partly verified
TC-13	FR-06	Edit and delete from a report	Use the record menu in a report	Record updated or removed	[To be recorded]	Not recorded
TC-14	FR-09	Meeting reminder e-mail	Wait for the schedule time	Reminder e-mail received	Schedules are inactive	Not executed
TC-15	FR-10	Access with a Read-profile user	Log in as a user with the Read profile	Only view access	No additional users configured	Not executed

6.3 Debugging Deluge Scripts

Three techniques were used. First, every form workflow writes info messages, and after a submission the form shows a “View Log Details” link (Figure 8.1) through which the messages of the workflows that ran can be read; this confirms that a workflow was triggered. Second, the values on the dashboard were compared with the number of records in each report, which confirmed the KPI cards and exposed the “Unknown” bar in the per-member chart. Third, the exported MeetMind.ds file was read to check the trigger, status and expression of every workflow and dashboard component.

The third technique found the defects that are not visible on screen: the two pairs of workflows on Meetings and Action_Items that share the same trigger, the KPI strip panels without criteria, the chart that reads from the sample form, the inactive schedules, and the UTC time zone. These defects are recorded in Table 6.2. They have not been fixed, because the application was not modified while this report was prepared.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	In the dashboard KPI strip, ACTIVE MEMBERS and TOTAL MEMBERS always show the same value, and COMPLETED ACTIONS shows the total number of action items.	ACTIVE MEMBERS and TOTAL MEMBERS both use thisapp.Team_Members.Member_Name.count; COMPLETED ACTIONS uses thisapp.Action_Items.Action_field.count with no criteria.	None. Proposed: add criteria Member_Status == “Active” and Action_Status == “Completed”.	Open
BUG-02	The dashboard chart “Daily New Registrations” shows data unrelated to meetings.	The chart reads New_Registrations from the sample Data_Source form.	None. Proposed: remove the chart or replace it with a MeetMind chart; remove the sample form and its menu items.	Open
BUG-03	Duplicate workflows run on every meeting and action-item submission.	Meeting_Creation_Notification and Meeting_Update_Confirmation (and the two Action_Items workflows) are all triggered on add or edit.	None. Proposed: set the creation workflows to On Add and the update workflows to On Edit.	Open
BUG-04	The Action Items by Team Member chart shows “Unknown”.	The Team_Members lookup is optional and was left empty; the assignee was typed in the separate Assigned_to text field.	None. Proposed: make Team_Members mandatory and remove or derive Assigned_to.	Open
BUG-05	The minutes titled “Client Feedback Meeting” are linked to the meeting “Construction Update”.	Meeting_Title and Meeting_Date in Meeting_Minutes are entered by hand and are not filled from the selected meeting.	None. Proposed: fill both fields from the lookup with a Deluge On User Input script and correct the record.	Open
BUG-06	The application time zone is UTC instead of Indian Standard Time.	Time zone set to “UTC” in the application settings; the schedules also use UTC, so 07:10 UTC corresponds to 12:40 IST.	None. Proposed: set the time zone to Asia/Kolkata.	Open

6.5 User Acceptance Testing

Formal user acceptance testing was not conducted. The application was tested only by the project team, as described in Section 6.1. User acceptance testing with members of a real team, using separate user accounts, is listed as future work.

Table 6.3: User Acceptance Feedback

User Role	Tested	Feedback	Action Taken
Meeting Organiser	N / Not Implemented	Formal user acceptance testing was not conducted.	—
Team Member	N / Not Implemented	Formal user acceptance testing was not conducted.	—

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. All work reported here was done in the Development environment, and the application was not moved to Stage or published to Production; Stage and Production deployment is therefore listed as N / Not Implemented. The planned deployment strategy is to fix the defects in Table 6.2 in Development, move the application to Stage for user acceptance testing with real team members, and then publish it to Production with version notes. The exported MeetMind.ds file (version 1.0, generated on 06-Nov-2026) serves as a backup of the current definition. Table 7.1 records the current state of the deployment checklist.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (√)
1	All test cases executed in Development / Stage	✗ (basic manual testing only)
2	Sample/test data removed or isolated	✗ (sample Data_Source form and test records present)
3	Roles, permissions and users configured in Production	✗ (default profiles only)
4	Connections authorised in Production	Not applicable (no connections)
5	Schedules and workflows enabled	Partly (form workflows active; schedules inactive)
6	Application published and URL verified	✗ (Production not used)
7	Version / release notes recorded	✗ (only the .ds export, version 1.0)

7.2 Performance Optimisation

No performance measurements were taken and no specific optimisation was applied. Table 7.2 records the one design decision with a bearing on performance, described qualitatively rather than as a measured result.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Built-in form-data aggregates (count, count with criteria) in the page panels instead of scripts that fetch records	User Engagement Dashboard	Not applicable (designed this way from the start)	Counts computed by the page components	Not benchmarked

Optimisation Applied	Component	Before	After	Improvement
Other optimisations (report column limits, filters, indexed criteria)	Reports	All fields shown	Not applied	Not measured

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account sign-in for application access	Whole application	Access by people who are not signed in and not shared on the application
Default permission profiles (Administrator, Read, Write, Customer)	Application share settings	Users receiving more permissions than intended (once users are added)
Personal-data flag on Member_Email	Team_Members form	Exposure of personal data
Mandatory Meetings lookup	Meeting_Minutes form	Minutes saved without a related meeting
Custom profiles and record-level criteria — N / Not Implemented	All forms and reports	Unauthorised data access between users
OAuth 2.0 scopes for API access — N / Not Implemented	REST API	Not applicable (API not used)

7.4 Governance and Administration

Governance in this project is mostly about naming, backup and ownership. Forms and reports follow a readable naming pattern (for example Meeting_Minutes and Meeting_Minutes_Report), although some link names are shortened or unclear (Action_field, Action_De), some display names carry extra spaces (“Meeting Date”, “Meeting Time ”), and the sample Data_Source form and its report are still in the application. The application definition was exported as MeetMind.ds on 06-Nov-2026; no source code repository is maintained. The application owner is responsible for maintenance, the trial subscription and future user management.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows one meeting through the application. In step 1, the organiser records a meeting through the Meetings form; after submission the form shows the "View Log Details" link of the form workflows. In step 2, minutes are recorded against a meeting, and the Zia Keyword Extractor fills the Extracted_Keywords field from the meeting notes. In step 3, the action items agreed in the meeting are recorded and tracked in the Action_Items Report, and the dashboard (Figures 5.8 and 5.9) shows the updated counts, charts and board.

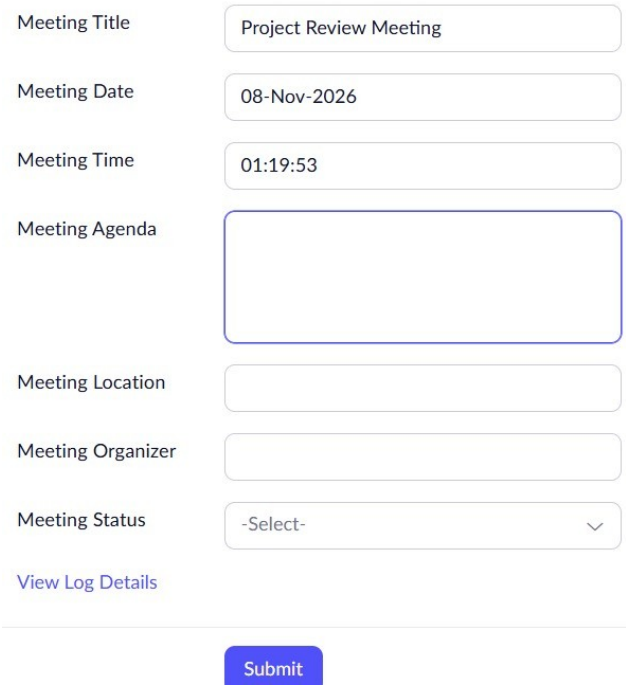


Figure 8.1: Scenario Step 1 – Meeting Recorded through the Meetings Form

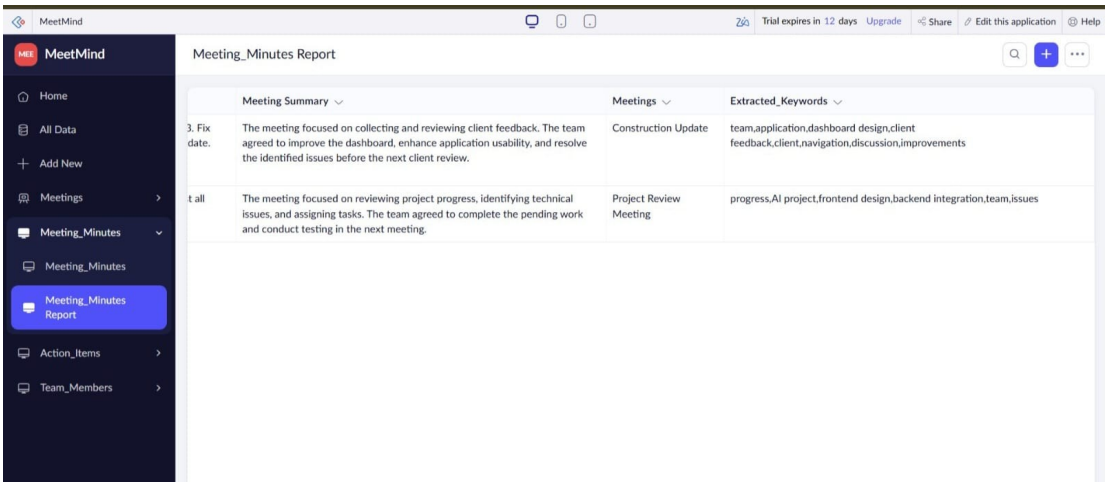


Figure 8.2: Scenario Step 2 – Keywords Extracted by Zia from the Meeting Notes

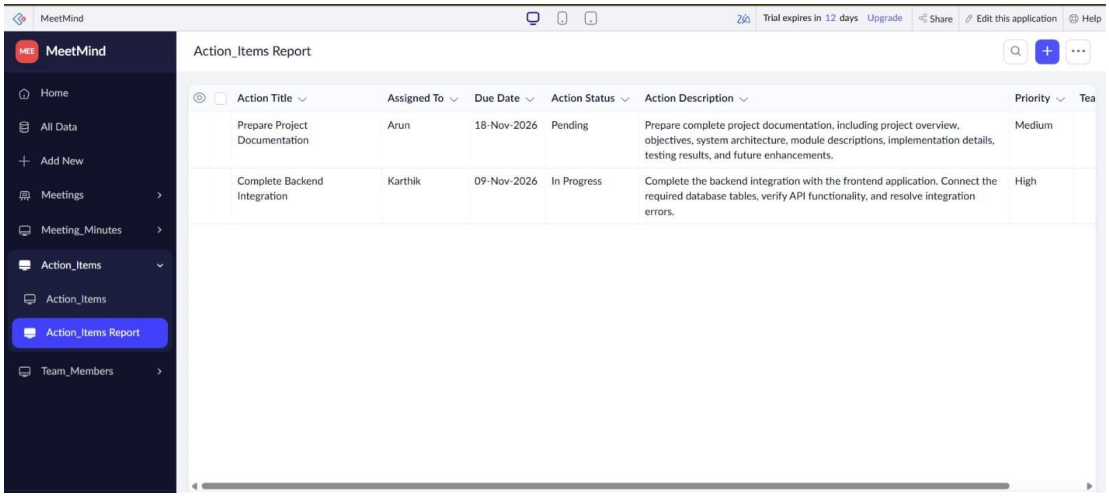


Figure 8.3: Scenario Step 3 – Action Items Tracked in the Action_Items Report

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Data model of four linked forms	Sections 4.2, 4.3, 5.2	Four MeetMind forms with two lookup fields; one mandatory lookup. The sample Data_Source form is still present.
2. AI keyword extraction from meeting notes	Section 5.9	Achieved: Zia Keyword Extraction field filled keywords for both test records (Figure 8.2, TC-04).
3. Reports and dashboard	Sections 5.3, 5.4	Four MeetMind list reports and a dashboard page; KPI cards match the reports (TC-07), but the KPI strip and two charts have defects (BUG-01, BUG-02, BUG-04).
4. Form and schedule workflows	Sections 5.5, 5.6	Partly achieved: six form workflows write log messages; the two reminder schedules are configured but inactive and untested (TC-14).
5. Document access, testing and defects	Sections 4.6, 5.8, 6, 7.3	Default profiles documented; basic manual testing recorded; six open defects logged; custom profiles not configured.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording a meeting	Details spread across e-mails and chat messages	One Meetings record with date, time, agenda, location, organiser and status	Single, structured record
Minutes and decisions	Typed late, inconsistent format	Meeting_Minutes form linked to the meeting; keywords extracted by Zia	Consistent and searchable

Parameter	Existing System	Proposed Application	Improvement
Task follow-up	Verbal agreement; owner and due date unclear	Action item with assignee, due date, priority and status	Traceable task status
Status reporting	Compiled by hand before each meeting	Dashboard with counts, charts, board and embedded reports	No manual compilation (subject to the open dashboard defects)

8.4 Limitations of the Low-Code Approach

The main limits met on Zoho Creator were these. First, the dashboard is assembled from UI Builder panels whose values are set one by one through expressions and criteria; a missing criterion is not flagged by the platform, which is how the KPI strip came to show identical or unfiltered counts (BUG-01). Second, a lookup field does not by itself copy values such as the title and date from the parent record; this needs a Deluge On User Input script, and without it the minutes could be linked to the wrong meeting (BUG-05). Third, the Zia Keyword Extraction field returns a fixed comma-separated list that cannot be tuned, and it identifies topics but does not produce summaries or action items.

Fourth, the application and its schedules run in the time zone set in the application settings, which was left at UTC, so schedule times must be converted for Indian users (BUG-06). Fifth, the application was built on a trial edition with a limited period, and it depends on one vendor’s platform, so moving it elsewhere would mean rebuilding it, although the exported .ds file helps with backup. Finally, the sample form and report that Zoho Creator creates with a new application remain in the application until they are removed by hand (BUG-02).

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the no-code to pro-code spectrum, almost all of MeetMind is no-code configuration: the four forms and their lookups, the five list reports, the dashboard page with its panels, charts and board, the Zia Keyword Extraction field, the reminder schedules with their send-email action, and the default profiles. The low-code part is small: six Deluge scripts that write info messages. There is no pro-code element. Moving towards low-code would add value in the near term—for example, Deluge On User Input scripts to fill the minutes from the selected meeting and e-mails to the assignee when an action item is created. A move to pro-code, for example with Zoho Catalyst, would make sense for AI-generated meeting summaries or automatic detection of action items in the notes, which go beyond what the ready-made AI fields provide.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

MeetMind shows that the core of a meeting-minutes and action-tracking application can be built on Zoho Creator with four linked forms, list reports, one dashboard page, six short form workflows and a ready-made Zia AI field. Meetings, minutes, action items and team members are now stored in one place, the key topics of each set of notes are extracted automatically, and the dashboard shows the number of meetings, action items and completed tasks without manual counting.

The report also records what is still open: the six defects in the defect log, the inactive reminder schedules, the absence of users and custom profiles, the lack of Stage and Production deployment, and the limited, manual testing. These are the first items of future work.

9.2 Future Enhancements

- Fix the six open defects: correct the KPI strip criteria, remove the sample Data_Source form and chart, separate the On Add and On Edit workflows, make the Team_Members lookup mandatory, fill the minutes from the selected meeting, and set the time zone to Asia/Kolkata.
- Activate and test the meeting-reminder schedules, and send e-mails to assignees when an action item is created or its due date is near.
- AI-generated meeting summaries and automatic detection of action items in the meeting notes, possibly with a Catalyst service.
- A Kanban report of action items grouped by Action Status and filtered reports for each team member.
- Add users with custom profiles and record-level criteria, so that each team member sees and updates only their own action items.
- Deploy through the Stage and Production environments with version notes.
- End-to-end testing of forms, workflows, schedules and access controls, followed by user acceptance testing with real team members.

9.3 Learning Outcomes and Career Relevance

The project gave practice in data modelling with lookup fields, list reports, dashboard design with the UI Builder, form and schedule workflows, Deluge scripting and the use of a Zia AI field. Reviewing the exported application definition also showed the value of checking every component’s configuration rather than relying only on what is visible

on screen. These skills relate directly to the roles of low-code developer, solutions consultant and automation engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Application MeetMind; ten menu sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Four forms use eight field types: Single Line, Date, Time, Multi Line, Drop Down, Lookup, Email, Keyword Extraction (AI)	5.2	Tables 4.2–4.5
II	Lookup field (relationship between forms)	Y	Meeting_Minutes: Meetings; Action_Items: Team_Members	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	N	Not used	4.2	—
II	Field validations / layout rules	P	Mandatory Meetings lookup; no layout rules	5.2	Table 4.3
II	List report with filter, grouping, custom action	P	Five list reports; no filter, grouping or custom action	5.3	Figures 5.5–5.7
II	Kanban / Calendar / Timeline / Map report	N	Not created	5.3	—
II	Pivot table / chart report	N	Charts are embedded in the dashboard page instead	5.4	Figure 5.8
II	Page / dashboard built with UI Builder	Y	User Engagement Dashboard	5.4	Figures 5.8, 5.9
II	Deluge – form events (validate / success / field actions)	P	On Success workflows with info statements only	5.5	Code 5.1–5.4
II	Deluge – functions, fetch, loops, collections	N	Not used	5.5	—
II	Email / in-app notifications	P	Send-email action in schedules (inactive)	5.6	App. C.7, C.8
II	Form workflow	Y	Six form workflows	5.6	Table 5.3
II	Scheduled workflow	P	Two schedules configured, status inactive	5.6	Table 5.3
II	Approval process	N	Not used	5.6	—
II	Blueprint	N	Not used	5.6	—
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	P	Basic manual testing; log link; .ds review	6	Table 6.1
III	Integration with a Zoho service	P	Zoho Mail through inactive schedules	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	N	Not Implemented	5.7	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Webhook or Creator REST API	N	Not Implemented	5.7	—
III	Users, roles and permission profiles	P	Default profiles and CEO role only	5.8	—
III	Record-level access criteria / customer portal	N	Not Implemented	5.8	—
III	Zia / AI feature	Y	Keyword Extraction field on Meeting_Notes	5.9	Figure 8.2
III	Analytics (charts / Zoho Analytics)	Y	KPI cards, charts and board on the dashboard page	5.4, 5.9	Figures 5.8, 5.9
IV	Performance optimisation applied & measured	N	Not measured; see Table 7.2	7.2	Table 7.2
IV	Deployment across environments	N	Development only	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	P	Sign-in, default profiles, personal-data flag	7.3	Table 7.3
IV	Governance (audit trail, administration)	P	.ds export; naming review	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	P	Phone and tablet layouts configured; no portal	5.11	—

Total features implemented: 8 fully and 11 partially / 32

Y = implemented; P = partially implemented; N = not implemented.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The scripts below are the complete Deluge code of the application, taken from the exported MeetMind.ds file. Sections C.7 and C.8 show the configuration of the two schedule workflows, whose send-email action is a built-in workflow action rather than a custom script. No source code repository is maintained for this project.

C.1 Meetings – Meeting_Creation_Notification (DS-01)

```
// Script ID : DS-01
// Location  : Meetings -> Form workflow "Meeting_Creation_Notification"
//             -> On Add or Edit -> On Success
info "MeetMind: Meeting form submitted successfully.";
info "Meeting details have been saved.";
```

C.2 Meetings – Meeting_Update_Confirmation (DS-02)

```
// Script ID : DS-02
// Location  : Meetings -> Form workflow "Meeting_Update_Confirmation"
//             -> On Add or Edit -> On Success
info "MeetMind: Meeting record updated successfully.";
info "Meeting changes have been saved.";
```

C.3 Meeting_Minutes – Meeting_Minutes_Submission (DS-03)

```
// Script ID : DS-03
// Location  : Meeting_Minutes -> Form workflow "Meeting_Minutes_Submission"
//             -> On Add or Edit -> On Success
info "MeetMind: Meeting minutes submitted successfully.";
info "Meeting minutes record has been saved.";
```

C.4 Action_Items – Action_Item_Creation_Notification (DS-04)

```
// Script ID : DS-04
// Location  : Action_Items -> Form workflow
"Action_Item_Creation_Notification"
//             -> On Add or Edit -> On Success
info "MeetMind: New action item submitted successfully.";
info "Action item has been recorded.";
```

C.5 Action_Items – Action_Item_Status_Update (DS-05)

```
// Script ID : DS-05
// Location  : Action_Items -> Form workflow "Action_Item_Status_Update"
//             -> On Add or Edit -> On Success
info "MeetMind: Action item record updated successfully.";
info "Please verify the latest task status.";
```

C.6 Team_Members – Team_Member_Registration (DS-06)

```
// Script ID : DS-06
// Location  : Team_Members -> Form workflow "Team_Member_Registration"
//             -> On Add or Edit -> On Success
info "MeetMind: New team member registered successfully.";
info "Team member record has been saved.";
```

C.7 Schedule Upcoming_Meeting_Reminder (form-based, inactive)

```
// Schedule  : Upcoming_Meeting_Reminder | form = Meetings | status = inactive
// Start     : Meeting_Date at "07:10:00" | End: Meeting_Date
// Frequency : daily | Time zone: UTC
// Action    : send mail (on start)
from        : zoho.adminuserid
to          : [college e-mail address of a team member]
subject     : "Upcoming MeetMind Meeting"
message     : "Hello, This is a friendly reminder that you have an upcoming
               meeting scheduled in MeetMind. Please check your meeting
               details and make sure you are available on time. Kindly
               review the meeting agenda and prepare any necessary
               discussion points. Thank you, MeetMind Team"
attachments : view : All_Meetings as PDF
```

C.8 Schedule Upcoming_Meeting_Reminder1 (custom, inactive)

```
// Schedule  : Upcoming_Meeting_Reminder1 (display name
Upcoming_Meeting_Reminder)
// Status    : inactive
// Start     : "2026-11-05 07:10:00" | Frequency: daily | Time zone: UTC
// Action    : send mail (on start) - same sender, recipient, subject and
//             message as C.7, without attachment
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the MeetMind application in a current web browser after signing in with the Zoho account with which the application has been shared. The published URL and demo credentials are not available at the time of submission: [To be added]. On a phone or tablet, use the Zoho Creator mobile app or the same link in the browser.

D.2 Instructions for the Meeting Organiser

- Open Meetings → Meetings and enter the meeting title, date, time, agenda, location, organiser and status; then click Submit.
- After the meeting, open Meeting_Minutes → Meeting_Minutes. Select the meeting in the Meetings field (mandatory), enter the meeting title and date exactly as in the selected meeting, and enter the notes, decisions taken and summary; then click Submit. The Extracted_Keywords field is filled automatically from the notes.
- Open Action_Items → Action_Items to record each task agreed in the meeting, and select the responsible person in the Team_Members field.

D.3 Instructions for the Team Member

- Open Action_Items → Action_Items Report to see the action items, their due dates and priorities.
- Open an action item, choose Edit, update the Action Status (Pending, In Progress, Completed or On Hold) and click Update.

D.4 Instructions for the Administrator

- Register team members through Team_Members → Team_Members and keep their role, department and status up to date.
- Use the four MeetMind reports to check and correct records; use Edit, Duplicate or Delete from the record menu when needed.
- Open Home to view the User Engagement Dashboard. Note that the KPI strip, the Daily New Registrations chart and the per-member chart are affected by open defects (Table 6.2).
- Ignore the All Data and Add New menu items, which belong to the sample Data_Source form and are not part of MeetMind.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	**SANSHIA J**	**Register No.**	**2116240701471**
Project Title	MeetMind – AI-Assisted Meeting Minutes and Action Tracker	Date	

Table E.1: Evaluation Rubric for the Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.

Level	% of Max. Marks	Descriptor
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**SANSHIA J**	**Register No.**	**2116240701471**
Department / Section	Computer Science and Engineering / CSE	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	**Question / Topic Asked**	**Unit**	**Response (E / G / S / NI)**	**Remarks**
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	**Max. Marks**	**Marks Awarded**
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____